OFFICIAL CURRICULUM HANDBOOK • CADRE TRACK: IMTC (INTEGRATED METEOROLOGICAL TRAINING TRACK)

Synoptic Meteorology Chart Analysis & INSAT-3DS Interpretation Guide

Comprehensive Field Guide on Surface Observations, Isobar Plotting, Frontal Dynamics, Water Vapor Channels, and WMO METAR Coding

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1. Surface Synoptic Observation & Weather Chart Plotting

Synoptic weather analysis forms the empirical bedrock of operational meteorological forecasting. Certified Meteorological Officers plot surface station models at standard synoptic hours (00, 03, 06, 09, 12, 15, 18, 21 UTC) according to World Meteorological Organization (WMO) Technical Regulation No. 49.

Synoptic Symbol	WMO Code Element	Plotting Standard	Diagnostic Significance
Isobars	Mean Sea Level Pressure (MSLP)	Solid curves at 2 hPa intervals	Pressure gradient, geostrophic wind direction
Station Circle	Cloud Cover (N in Okta)	Shaded 0/8 (Clear) to 8/8 (Overcast)	Boundary layer saturation and cloudiness
Wind Barb	Direction (dd) & Speed (ff)	Feathers: Half (5kt), Full (10kt), Flag (50kt)	Surface momentum advection
Present Weather (ww)	Precipitation Code (00-99)	International WMO standard glyphs	Active convection, fog, thunderstorm, rain

2. INSAT-3DS Geostationary Meteorological Sounder Channels

India's third-generation dedicated meteorological satellite INSAT-3DS positioned at 74°E and 93.5°E provides continuous 15-minute rapid scan multispectral imagery across visible, shortwave IR, thermal IR, and upper-tropospheric water vapor channels.

Water Vapor Band (6.7 µm): *Senses moisture distribution in the 300-600 hPa layer. Dark dry bands indicate descending stratospheric air in upper-level jet streaks, while bright white cloud shields delineate active deep monsoon depressions.*